

Avnish Pal

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Experience

Software Engineer (Contractor)

Jul 2024 – Present

Autodesk

Singapore

- Engineered a robust offline caching mechanism for a JWT-based authorization system, enabling Autodesk products to perform compliance checks without active internet connectivity.
- Implemented fault-tolerant state management for product authorization workflows, ensuring system integrity during machine restarts and hibernation through an enhanced event-based architecture.
- Resolved critical pipeline failures and upgraded Qt framework across Linux and MacOS environments for legacy products, improving development reliability and ensuring continued compatibility.

Software Quality Engineer

Jul 2021 – Jun 2024

Red Hat

Pune, India

- Designed and deployed automated CI/CD pipelines for cloud-native product certification, substantially reducing testing cycles while increasing validation coverage.
- Created a Slack-integrated notification system that provides real-time visibility into test results, improving team communication and accelerating issue resolution.
- Built comprehensive multi-cloud certification testing infrastructure for RHEL OS across AWS, Azure, and GCP, ensuring consistent product quality across diverse cloud environments.

Software Engineer

Sept 2019 – Jun 2021

Taiyo AI

Delhi, India

- Developed a predictive stock analysis platform using Python and machine learning frameworks that delivered actionable investment insights from large financial datasets.
- Engineered a distributed web scraping system that collected and processed economic indicators from numerous sources to enable effective supply chain risk assessment.
- Implemented RESTful APIs to deliver real-time commodity market insights, supporting high-volume data requests with robust performance.

Projects

Chat with Document | Python, Streamlit, LangChain, Ollama, Generative AI

- Developed a web application enabling users to upload documents and interact with them using natural language queries through integration with Ollama's self-hosted large language models.

DataWiz | Dash, Django, Web Development

- Created a data science prototype application using Django and Dash that combines data visualization and machine learning components into a unified workflow for rapid prototyping and quick data exploration.

Pull Request Analysis Tool | Grafana, Python, Data Analytics

- Built a Python application that extracts and processes GitHub Pull Request data to generate meaningful development metrics and visualize team performance through Grafana dashboards.

Education

Bachelor of Technology (Information Technology)

Guru Gobind Singh Indraprastha University

Delhi, India

Skills

Programming Languages: Go, Python, C++, TypeScript, JavaScript, SQL, Bash, Java, Dart

Libraries & Frameworks: ReactJS, AngularJS, SvelteJS, Streamlit, LangChain, scikit-learn, pandas, TensorFlow, PyTorch, pytest, Dash, Django, Flask, Selenium, Cypress, Apache Airflow, Flutter, Cobra CLI, Tekton Pipelines

Technologies: Kubernetes, OpenShift, OpenStack, Podman, Docker, Jenkins, Git, GitHub, GitLab, PostgreSQL, MySQL, SQLite, MongoDB, Grafana

Platforms: GNU/LINUX, AWS, GCP, IBM Cloud, Terraform, Ansible

Languages: English, German